Black Hole–Wormhole Structure and Wormhole Collapse in PPC Gravity

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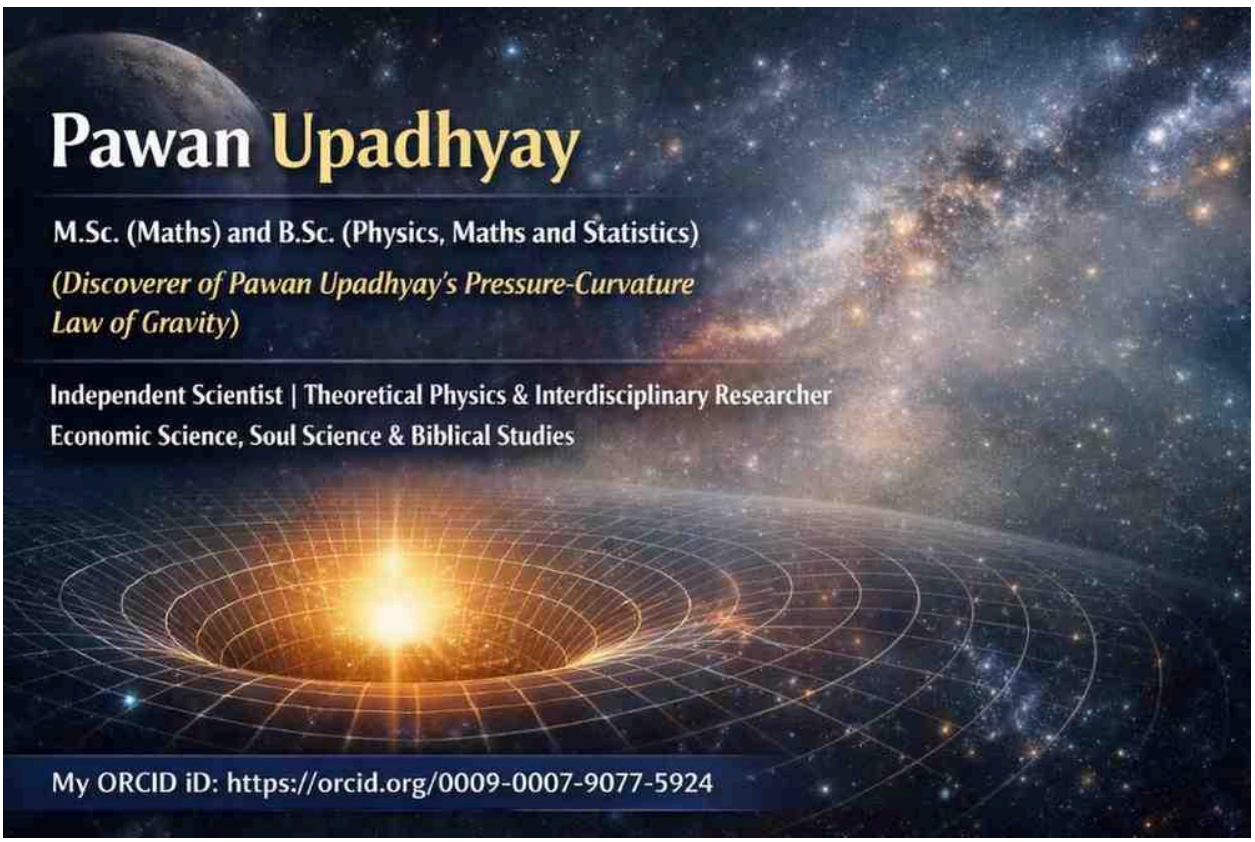

Abstract

This paper examines the possible relationship between **black holes, wormholes, and inter-universe connectivity** within the framework of the **Pawan Upadhyay Pressure–Curvature Law of Gravity (PPC Gravity)**.

Initially, black holes may appear capable of forming wormhole tunnels that connect distant regions of spacetime or separate universes. Such a structure can be conceptually represented as:

$$Universe_A \rightarrow BlackHole_A \rightarrow Wormhole \rightarrow BlackHole_B \rightarrow Universe_B$$

However, PPC gravity predicts that extremely high **gravitational pressure and energy density** near black holes cause wormhole throats to collapse. As a result, stable wormhole bridges cannot persist under extreme pressure conditions.

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1. Introduction

Black holes represent regions of extremely strong gravitational fields. In many theoretical models, wormholes have been proposed as spacetime tunnels connecting distant parts of the universe or even different universes.

The **PPC Law of Gravity** introduces an alternative interpretation of gravitational phenomena through the concept of **gravitational pressure generated by spacetime energy density**.

The fundamental equation of PPC gravity is

$P_g = \omega E_d$

where

- P_g = gravitational pressure
- E_d = spacetime energy density
- ω = equation of state parameter or Pressure Energy density coupling parameter or Pressure Curvature Coupling Parameter.

This relationship suggests that spacetime curvature increases as energy density increases.

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2. Pressure Gradient and Gravitational Force

In continuum physics, forces generated by pressure fields arise from the pressure gradient:

$$\vec{F} = -\nabla P_g$$

The negative gradient indicates that the force acts toward regions of lower pressure.

Substituting the PPC relation

P_g=ωE_d

we obtain

$$\nabla P_g = \omega \nabla E_d$$

Thus the gravitational force density becomes

$$\vec{F} = -\omega \nabla E_d$$

This expression shows that gravity emerges from **spatial variations in energy density**.

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3. Extreme Conditions Near Black Holes

Near the event horizon of a black hole, spacetime energy density becomes extremely large.

In PPC gravity, this leads to a limiting condition:

$P_g = E_d$

which corresponds to

$\omega = 1$

Under this condition

$$\nabla P_g = \nabla E_d$$

and the gravitational force density becomes

$$\vec{F} = -\nabla E_d$$

This represents an extremely strong inward pressure field.

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4. Possible Black Hole–Wormhole Connection

Some theoretical models suggest that black holes might connect to wormhole tunnels that lead to other universes.

The conceptual structure can be expressed as

$$Universe_A \to BlackHole_A \to Wormhole \to BlackHole_B \to Universe_B$$

In this scenario:

- **Black Hole A** acts as an entrance
- the **wormhole tunnel** connects two regions of spacetime
- **Black Hole B** acts as an exit in another universe.

Such a structure would represent a **cosmic bridge between universes**.

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5. Wormhole Stability in PPC Gravity

For a wormhole to remain open, the geometry near the wormhole throat must resist gravitational collapse.

However, in PPC gravity gravitational pressure is proportional to energy density:

P_g=ωE_d

Therefore, regions of extremely high energy density produce extremely high gravitational pressure.

This pressure acts inward on the wormhole throat.

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6. Wormhole Collapse Condition

If the gravitational pressure exceeds the structural stability of the wormhole throat, collapse occurs.

The collapse condition may be written as

P_g > P_{crit}

where P_{crit} represents the pressure required to maintain the wormhole throat.

Near black holes:

$P_g = E_d$

Since E_d becomes extremely large, the collapse condition is easily satisfied:

$E_d > P_{crit}$

Thus the wormhole throat becomes unstable and collapses.

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7. Physical Interpretation

Within PPC gravity, wormholes behave like spacetime structures subjected to intense gravitational pressure.

Extremely high pressure produces:

- strong inward curvature
- compression of spacetime geometry
- instability of the wormhole throat.

As a result, the wormhole tunnel closes.

This implies that wormholes cannot remain stable in environments of extremely high energy density.

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8. Implications for Inter-Universe Connectivity

Although theoretical models suggest the possibility of black hole–wormhole connections, PPC gravity predicts that such structures are unstable.

Instead of forming permanent bridges between universes, wormholes are expected to collapse under the extreme pressure conditions near black holes.

Thus black holes are unlikely to serve as stable gateways between universes.

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9. Conclusion

The **PPC Law of Gravity** describes gravity as gravitational pressure generated by spacetime energy density:

$P_g = \omega E_d$

Near black holes the limiting condition

$P_g = E_d$

produces extremely strong pressure fields and large pressure gradients.

While theoretical models suggest that black holes could connect to wormholes, the PPC framework indicates that the resulting high pressure and energy density would destabilize the wormhole throat.

Consequently, wormholes collapse before forming stable cosmic bridges, making inter-universe connections through black holes unlikely within PPC gravity.

Prediction Comparison: PPC Gravity vs General Relativity

Phenomenon	General Relativity (GR)	PPC Law of Gravity
Fundamental description of gravity	Gravity arises from **spacetime curvature** produced by the energy–momentum tensor	Gravity arises from **gravitational pressure generated by spacetime energy density**
Core relation	Einstein field equation	PPC pressure–energy relation
Origin of gravitational force	Geodesic motion in curved spacetime	Pressure gradient force
Role of energy density	Energy density contributes to curvature through Field equation.	Energy density directly determines gravitational pressure

Black hole interior	Predicts a **singularity** with infinite density	Predicts an **extreme pressure–energy density state**
Wormhole stability	Wormholes theoretically possible with exotic matter	Wormholes **collapse due to high pressure and high energy density**
Hawking radiation mechanism	Quantum particle creation near the event horizon	Radiation generated by **extreme pressure gradients near the horizon**
Behavior near event horizon	Geometry dominated by spacetime curvature	Region where **gravitational pressure approaches energy density**
Gravitational force interpretation	Motion along curved spacetime geodesics	Motion driven by **pressure gradients in spacetime energy density**
Extreme gravity regime	Curvature becomes dominant	Pressure–energy density equality condition governs dynamics

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Key PPC Prediction

A central prediction of PPC theory is the **pressure–energy density equality condition near black holes**:

$P_g = E_d$

This condition leads to:

- extremely large pressure gradients
- strong inward spacetime compression
- **instability of wormhole throats**

Thus PPC gravity predicts that **stable wormhole tunnels cannot exist near black holes**.

P_g ➡️ P_g (Max)

E_d ➡️ E_max

P_g ~ E_d because ω ~ 1

In PPC Law of Gravity, ω is variable, not constant. The numerical value of omega (ω) is approximate, not absolute. Spacetime itself is not absolute.

Explanation Note on Variable Equation-of-State Parameter in PPC Law of Gravity

In Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity), the gravitational pressure field is defined as:

$$P_g = \omega E_d$$

where:

- P_g = gravitational pressure field,
- E_d = energy density,
- ω = equation-of-state parameter.

The gravitational field force density is defined by:

$$F = -\nabla P_g$$

Substituting the pressure relation:

$$F = -\nabla(\omega E_d)$$

Using the vector product rule for gradients:

$$F = -\nabla(\omega E_d) = -(\omega \nabla E_d + E_d \nabla \omega)$$

This equation shows that the PPC gravitational force density contains two separate physical contributions:

1. **Energy Density Gradient Contribution**

$$-\omega \nabla E_d$$

This term represents the force density generated due to variation in energy density across space.

2. **Equation-of-State Gradient Contribution**

$$-E_d \nabla \omega$$

This additional term appears because the equation-of-state parameter ω is treated as a variable quantity in PPC Law of Gravity.

If ω changes with spacetime coordinates, then:

$$\nabla\omega \neq 0$$

and the second term contributes to the total force density field.

However, when a fixed numerical value of ω is substituted, ω becomes constant locally. Therefore:

$$\nabla\omega = 0$$

and the force density equation simplifies to:

$$F = -\omega\nabla E_d$$

Thus, the generalized PPC formulation allows a dynamical equation-of-state parameter, while the constant-ω approximation naturally removes the coupling-gradient contribution.

This interpretation provides a more flexible gravitational framework in which both energy density variations and equation-of-state variations can influence the gravitational force density field.

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3. Behavior Near the Black Hole Interior

As matter collapses toward the center of a black hole:

- energy density increases
- gravitational pressure also increases.

According to PPC gravity, the system approaches a limiting state:

$$P_g = E_d$$

This represents a **maximum pressure–energy density condition**.

Instead of producing an infinite-density point, the interior may reach an **extreme pressure equilibrium state**.

4. Replacement of the Singularity

In PPC gravity, the singularity may be replaced by a **finite but extremely dense pressure state**.

GR prediction

$$\rho \to \infty$$

PPC prediction

$$P_g \approx E_d$$

This means

- density becomes extremely large but **not necessarily infinite**
- gravitational pressure balances further collapse.

Thus the black hole interior may behave like a compressed spacetime core rather than a mathematical singularity.

5. Physical Picture of the PPC Interior

Inside the black hole:

1. Matter collapses and energy density rises.
2. Gravitational pressure increases according to

$$P_g = \omega E_d$$

3. When the equality condition is reached

$$P_g = E_d$$

spacetime enters an **extreme pressure regime**.

This regime produces:

- maximum spacetime compression
- extremely strong pressure gradients
- possible quantum effects such as radiation emission.

6. Implications of the PPC Interior Model

If PPC gravity is correct, the interior of a black hole could have several different properties compared with GR.

Possible consequences

• No mathematical singularity
• Interior dominated by **pressure–energy equilibrium**
• Extremely compressed spacetime core
• Modified behavior of Hawking radiation.

✔️ Summary

In **General Relativity**, the center of a black hole is a singularity where physical quantities become infinite.

In **PPC Law of Gravity**, gravitational pressure increases with energy density until reaching the extreme condition

$$P_g = E_d$$

This may prevent the formation of a true singularity and instead produce a **maximally compressed pressure-dominated spacetime core** inside the black hole.